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HIStore'y

An E-commerce website in NodeJS for
Partial fulfillment of the Requirements in
The Subject titled Advance Web Application

BSIT-S-2A

by

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INTRODUCTION

HIStore'y is **a full-stack web application built on Node.js** that serves as a dedicated online memorabilia store celebrating the life and legacy of **Michael Jackson** — the King of Pop. The platform provides enthusiasts, collectors, and fans worldwide with a curated marketplace to discover, browse, and purchase authentic memorabilia, limited-edition collectibles, apparel, and multimedia items inspired by one of the greatest entertainers in music history.

The name HIStore'y is a portmanteau of "HI" (referencing Jackson's signature exclamation and greeting to his fans), "Store" (the commercial purpose of the platform), and "Story" — paying homage to the storied career and enduring cultural impact of Michael Jackson. It is also a nod to his **1995 double album HIStory: Past, Present and Future, Book I**, underscoring the platform's mission to keep his legacy alive through meaningful collectibles.

The application is designed with two primary stakeholders in mind: customers who wish to browse and purchase products, and administrators who manage the store's inventory, orders, and user base. By leveraging modern web technologies — including Node.js for the backend, jQuery for dynamic front-end interactions, and a RESTful API architecture — HIStore'y delivers a seamless, secure, and feature-rich shopping experience.

**TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES - TAGUIG****BACKGROUND OF THE STUDY**

Michael Joseph Jackson (August 29, 1958 – June 25, 2009) remains one of the best-selling music artists of all time, with an estimated 350 million records sold worldwide. His influence on music, dance, fashion, and culture continues to permeate global popular culture more than a decade after his passing. Albums such as *Thriller* (1982), *Bad* (1987), *Dangerous* (1991), and *HIStory* (1995) are considered landmark works in the history of recorded music.

The global fanbase — known as MJ fans or "Moonwalkers" — spans multiple generations and continents. This passionate community actively seeks memorabilia, vintage merchandise, and collectible items that connect them to the artist. Despite this demand, the online marketplace for authenticated Michael Jackson memorabilia remains fragmented, with collectors relying on general-purpose platforms such as eBay or individual fan-operated websites that often lack robust e-commerce functionality.

**TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES - TAGUIG****PROBLEM STATEMENT**

The existing landscape for Michael Jackson memorabilia commerce presents several challenges:

- Lack of a dedicated, themed platform that combines community identity with professional e-commerce features.
- Absence of centralized inventory management tools for memorabilia sellers and store administrators.
- Limited transaction transparency, with inadequate email notifications and receipts for buyers.
- Poor user account management, with no role-based access control differentiating customers from administrators.
- Insufficient data visualization tools to help administrators understand store performance and sales trends.

OBJECTIVES

The HIStore'y project aims to:

- Develop a full-stack Node.js web application with a RESTful API backend for managing products, transactions, and users.
- Implement role-based authentication and authorization using JWT tokens to secure admin and customer routes.
- Provide a dynamic jQuery-powered front-end with DataTables for data management, pagination, and infinite scroll.

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- Integrate automated email notifications for order updates, complete with PDF receipts attached to confirmation emails.
- Deliver an intuitive, visually rich UI/UX design that reflects the artistry and cultural legacy of Michael Jackson.
- Incorporate data analytics dashboards featuring bar, line, and pie charts for administrative insights.

SCOPE AND LIMITATIONS

The scope of HIStore'y covers the following:

- Full CRUD operations for Products, Users, and Transactions accessible via a Node.js REST API.
- A jQuery/DataTables front-end for all CRUD interfaces with support for multiple file uploads (product images).
- JWT-based user registration, login, and session management with token storage on the Users table.
- Middleware-enforced route protection so only administrator roles can access sensitive CRUD endpoints.
- Automated transactional email notifications using a mail service, with PDF receipt generation on order updates.
- Three chart types (bar, line, pie) for the admin dashboard reflecting sales, revenue, and product data.
- Server-side and client-side search/autocomplete on the customer-facing homepage.



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- jQuery-based custom pagination and infinite scroll (distinct from DataTable pagination) on listing pages.

The following are explicitly outside this project's scope:

- Integration with live payment gateways (e.g., Stripe, PayPal); payments are simulated for this academic implementation.
- Mobile native applications; the platform is web-based and responsive but not a native app.
- Third-party authentication providers (e.g., Google OAuth, Facebook Login).

USE CASE DIAGRAM




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TECHNICAL STACK

Layer	Technology / Tool
Runtime & Backend	Node.js with Express.js framework
Database	MySQL / MariaDB (via Sequelize ORM or raw queries)
Authentication	JSON Web Tokens (JWT), bcrypt for password hashing
Front-End	HTML5, CSS3, jQuery, DataTables.js
File Uploads	Multer (multi-file middleware for Express)
Email Service	Nodemailer (SMTP) with Gmail or Mailtrap for dev
PDF Generation	PDFKit or Puppeteer for receipt generation
Charts	Chart.js (bar, line, pie charts)
Validation	jQuery Validation Plugin (client-side)
Search / Autocomplete	jQuery UI Autocomplete with API endpoint
Version Control	Git & GitHub

FUNCTIONAL REQUIREMENTS

Functional requirement	Actor	CRUD operation	Description	...
AUTHENTICATION & AUTHORIZATION				
User registration	Customer	Create	Register with email & password; JWT token generated and saved to users table	
User login	Both	Read	Authenticate via jQuery AJAX; JWT returned and stored client-side	
User logout	Both	Update	Invalidates active token on the users table	
Role-based route protection	Admin	Read	JWT middleware checks role on every request; non-admins get 403 on CRUD endpoints	
CUSTOMER MANAGEMENT				
Register / create profile	Customer	Create	New user record inserted with role = 'customer'; full name, address, phone optional	
View profile & order history	Customer	Read	Displays account info, active orders, and past transaction list	
Edit profile & change password	Customer	Update	AJAX form updates name, email, address, photo; password requires current password confirmation	
Deactivate account	Customer	Delete (soft)	Sets is_active = false; account hidden but data retained for order history	



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ADMIN — USER MANAGEMENT			
Create user account	Admin	Create	Admin can manually add any user with any role from the admin panel
List & search users	Admin	Read	DataTable with search/filter by name, email, role, or status
Assign / change user role	Admin	Update	Promotes or demotes users between 'customer' and 'admin' roles via AJAX
Activate / deactivate user	Admin	Update	Toggles is_active; deactivated users cannot log in
Delete user	Admin	Delete	Permanent removal after confirmation; soft-delete available as alternative
PRODUCT MANAGEMENT			
Add product listing	Admin	Create	Name, description, category, price, stock, condition, year; up to 5 images via Multer multi-upload
Browse & search products	Both	Read	Customer grid view with filters; admin DataTable view with sort/search
Search autocomplete	Customer	Read	jQuery UI Autocomplete on homepage queries Node.js API in real time
View product detail	Customer	Read	Image gallery, description, condition, stock status, and Add to Cart button
Edit product	Admin	Update	Edit all fields; add/remove individual images; update stock in real time via AJAX
Delete product	Admin	Delete (soft)	Soft-delete if transactions exist; hides from catalog but preserves order records
ORDER & TRANSACTION MANAGEMENT			
Place order / checkout	Customer	Create	Cart items submitted; transaction created with status = 'Pending'; confirmation email sent
View order list	Both	Read	Customers see own orders; admin sees all via DataTable with status/date filters
View order detail	Both	Read	Line items, quantities, unit prices, shipping address, current status
Update order status	Admin	Update	Pending → Processing → Shipped → Delivered; triggers email notification on every change
Send email notification	Admin	Update	Nodemailer sends status-change email; Shipped/Delivered emails attach a PDF receipt
Generate PDF receipt	Admin	Read	PDFKit generates itemised receipt with order details, totals, and HStore'y branding
Cancel order	Both	Update	Customer requests cancellation; admin approves and restores stock
Delete transaction	Admin	Delete	Permanent removal of cancelled or test orders after confirmation


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ADDITIONAL FEATURES			
Client-side form validation	Both	—	jQuery Validation Plugin on all CRUD forms; inline error messages without page reload
Custom pagination	Customer	Read	jQuery-based page navigation on product listing; uses limit/offset API params
Infinite scroll	Customer	Read	Auto-loads more products as user scrolls; distinct from DataTable pagination
Analytics dashboard — bar chart	Admin	Read	Monthly sales volume by product category via Chart.js
Analytics dashboard — line chart	Admin	Read	Revenue trend over the past 12 months via Chart.js
Analytics dashboard — pie chart	Admin	Read	Proportion of orders by status (Pending, Shipped, Delivered, Cancelled)
Multiple file upload	Admin	Create / Update	Multer middleware accepts up to 5 images per product; JPEG, PNG, WebP only
DataTables integration	Admin	Read	Server-side or client-side DataTables on all admin CRUD list views

DATABASE SCHEMA OVERVIEW

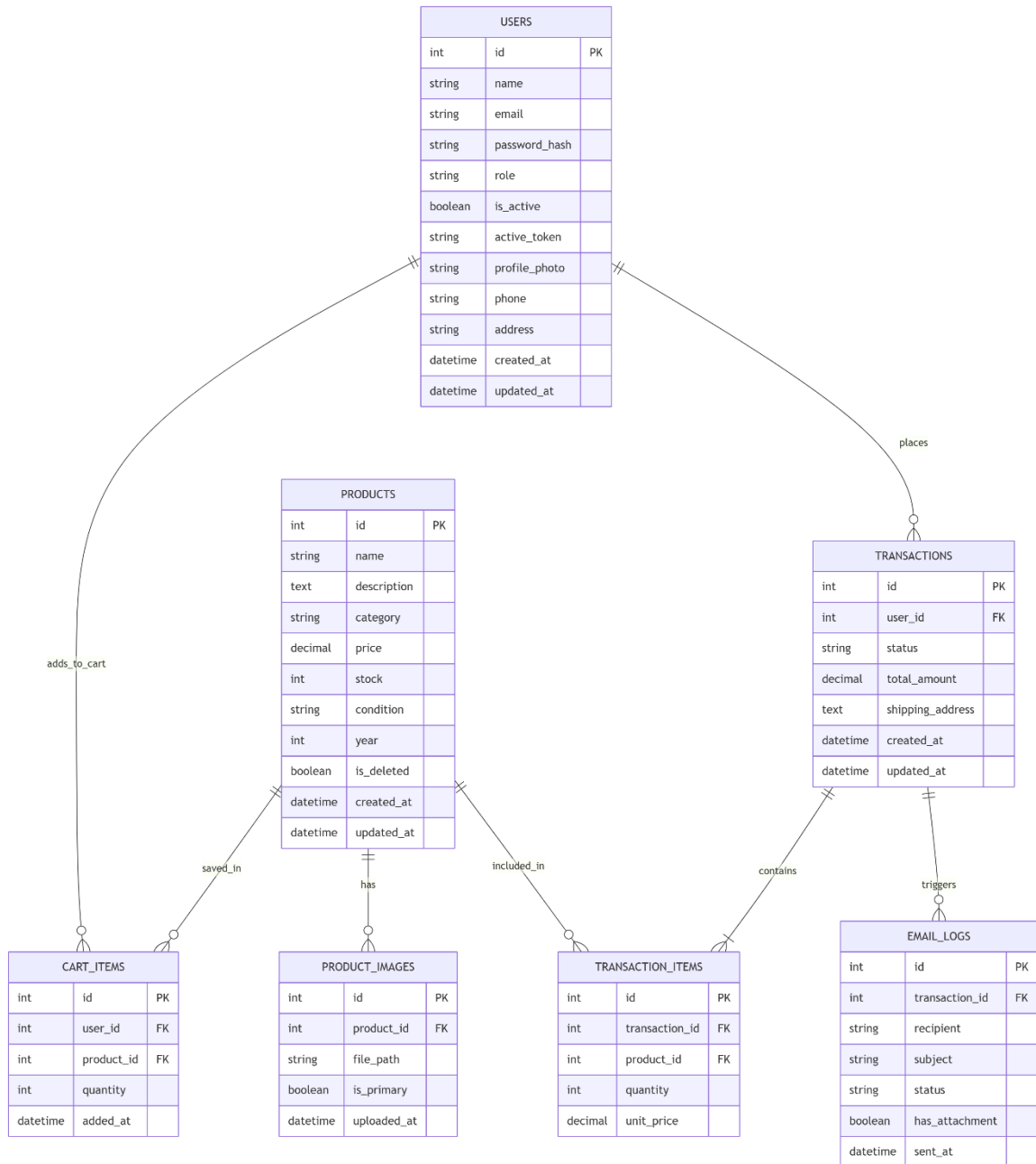
The following core tables constitute the HIStore'y data model:

Table	Key Columns & Purpose
users	id, name, email, password_hash, role (admin/customer), is_active, active_token, created_at
products	id, name, description, category, price, stock, condition, year, is_deleted, created_at
product_images	id, product_id (FK), file_path, is_primary, uploaded_at
transactions	id, user_id (FK), status, total_amount, shipping_address, created_at, updated_at
transaction_items	id, transaction_id (FK), product_id (FK), quantity, unit_price
email_logs	id, transaction_id (FK), recipient, subject, status, sent_at



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ENTITY RELATIONSHIP DIAGRAM



**TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES - TAGUIG****SECURITY CONSIDERATIONS**

- All passwords are hashed with bcrypt (salt rounds: 10+) and never stored in plaintext.
- JWT tokens are signed with a server-side secret key and include an expiration claim (exp).
- Route protection middleware validates the JWT on every sensitive API request, returning 401 Unauthorized if the token is missing, expired, or tampered with.
- Role enforcement middleware returns 403 Forbidden if a customer-role user attempts to access an admin-only endpoint.
- File upload validation restricts accepted MIME types to image formats (JPEG, PNG, WebP) and enforces a maximum file size limit.
- All database queries use parameterized statements to guard against SQL injection attacks.

HISTORE'Y – his story lives on